SQL Complete Guide

Version 1.0 - PrepAI Knowledge Base

1. Introduction

SQL (Structured Query Language) is used to manage and query relational databases.

2. Database Concepts

Database, Table, Row, Column, Primary Key, Foreign Key, Constraints.

3. SQL Commands

DDL (CREATE, ALTER, DROP), DML (INSERT, UPDATE, DELETE), DQL (SELECT), DCL (GRANT, REVOKE), TCL (COMMIT, ROLLBACK).

4. SELECT Queries

SELECT, WHERE, ORDER BY, GROUP BY, HAVING, LIMIT.

5. Joins

INNER JOIN, LEFT JOIN, RIGHT JOIN, FULL OUTER JOIN, SELF JOIN, CROSS JOIN.

6. Functions

Aggregate functions: COUNT, SUM, AVG, MIN, MAX. String and Date functions.

7. Subqueries

Single-row, multi-row, correlated subqueries.

8. Indexes & Views

Indexes improve query performance. Views provide virtual tables.

9. Normalization

1NF, 2NF, 3NF, BCNF and denormalization concepts.

10. Common Interview Questions

• Difference between DELETE, TRUNCATE and DROP?
• What is a Primary Key?
• What is a Foreign Key?
• Explain normalization.
• Difference between WHERE and HAVING.
• Types of JOINs.
• What are indexes?
• What is a view?
• What is a stored procedure?
• What is a transaction?

11. Practice Queries

Write queries for employee, student and sales databases using joins, group by, subqueries and aggregate functions.

12. Best Practices

Use indexes wisely, avoid SELECT *, write optimized joins, use transactions for data integrity.